Do Faithful Explanations Go with Accurate Forecasts? An Audit of Plan-Lift Explanations in Retail Forecasting

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Abstract. Retail forecasting models that see the price and promotion plan can explain each forecast as a baseline, what would sell with nothing planned, plus the lift each part of the plan adds. Merchants read that split as fact and plan on it. We set out eight checks on such explanations, fixed in advance: properties that hold by construction, sensitivity to randomised weights, stability across seeds, price direction, agreement with what sold, agreement with LightGBM, a plan read-back, and agreement with a measured cause. In generated store data with known effects, a world model whose explanation is built in passes C1 and C3 to C6, and its lifts match the truth (rank agreement 0.99) when plans are made at random. When merchants plan into high-demand weeks, it still passes them, forecasts more accurately, and overstates the total lift by a third (1.33 to 1.40 times) and the display's by 1.81 to 1.93 times; LightGBM overstates them too, and the two agree. Asked about a lasting change in the regular price, the world model answers with the temporary-cut response, -2.42 to -2.37 against a true -0.5, and only the measured cause finds it. The price-direction check catches every model blind to price but also flags LightGBM; an exploratory lever-by-lever comparison catches lifts given to the wrong lever; the randomisation check separates nothing. On Dominick's we find 54 spells in which a store's regular prices sat apart from its zone peers, most between April 1993 and April 1994. Over the 38 scored spells, prices 6% to 10% off their peers, the pooled effect on category units is -0.33 over eight weeks (95% interval -0.68 to 0.01) and -0.52 over the full spell (-0.81 to -0.23), in line with the elasticity near -0.3 the chain's experiments implied. These pooled intervals were added after a placebo at false start dates gave -0.26 (-0.72 to 0.19) and showed single episodes to be noise. *The audit of every saved model version from our companion paper on M5, Dominick's and Breakfast at the Frat, and the test of whether versions whose explanations pass more checks forecast better, are running and complete this draft.*

Keywords: retail forecasting; explanation; faithfulness; promotions; price response; difference in differences; world models.

1 Introduction

A retail forecast is used for more than its number. When a merchant plans a promotion, the question is not only how much will sell but how much of it the promotion will add, and whether a deeper price cut would add more. Forecasting models that see the plan can answer that directly: they forecast the item with the plan as made, and again with nothing planned, and the difference is the lift the plan adds. The world model and the lift model of our companion paper [34] are built this way: each forecast is a baseline under the nothing-planned plan plus a lift, and the lift is exactly zero when nothing is planned. The split is what a merchant reads ("the promotion added 40 units"), plans on, and is judged by.

Such an explanation is useful only if it can be trusted, and being built into the model does not make it right. The explanation literature separates two properties [1, 36]: an explanation is *faithful* if it reflects what the model does, and *correct* if it reflects what happens in the world. In molecular property prediction, MolSanity found the two come apart: attributions faithful to the model were often anti-aligned with the substructure that determines the property [2]. Retail has a property chemistry lacks. Some causes can be measured: Dominick's Finer Foods moved store prices on purpose in the early 1990s [3], so the response of sales to a price change can be estimated from the data and a model's stated response compared with it.

This paper audits plan-lift explanations in retail forecasting. It asks three questions, fixed in a plan before any result [35]: are the explanations faithful to the model, stable and sensible; do they match what sold and what a price change causes; and across model versions, do versions whose explanations pass more checks also forecast better?

Contributions.

1. A definition of the plan-lift explanation of a forecast that applies to any model that reads the plan, and eight checks on it (Figure 1), each with a pass rule fixed in advance.
2. A known-answer study: generated store data in which the effect of every part of the plan is known,

with failures planted on purpose, to measure which checks catch which failure before they are trusted on real data.

3. A ground truth on public retail data: 54 spells in six Dominick's categories where one store's regular prices sat apart from its zone peers, 38 of them scored, with their measured effect on sales by difference in differences, an event study, a placebo and a sensitivity analysis.
4. An audit of every saved world-model and lift-model version of the companion paper on M5, Dominick's and Frat, against LightGBM, and the test of whether faithful explanations go with accurate forecasts.

Findings. In generated data with known effects, explanations that pass every check against the model, the data and another model can still be wrong in size, when merchants plan into strong weeks, and wrong in kind, when a model answers a question about the regular price with what it learned from temporary cuts. The cheap price-direction check and a lever-by-lever comparison with a second model catch the cruder failures; the randomisation check does not test an explanation built into the model; and only a measured cause finds the error in kind (§4). On Dominick's, zone price deviations give a pooled measured response in line with the chain's own experiments, but imprecise, and single episodes are noise (§6).

2 Related work

What an explanation should satisfy. Interpretability lacks a single definition [38, 39]. Jacovi and Goldberg [1] separate faithfulness, whether an explanation reflects the model's reasoning, from plausibility, whether it convinces a person, and warn that evaluations often measure the second. Chen et al. [36] draw the related line between explanations true to the model and true to the data, and Janzing et al. [37] argue that feature relevance is a causal question about interventions on the model's inputs. Our on/off contributions are interventions of exactly that kind, made on the plan inputs the merchant controls.

Feature attributions and their failures. LIME [4] and SHAP [5], with its exact form for trees [6], are the most used attributions [7]. Saliency maps can barely change when a network's weights are randomised [8], move with input shifts the model ignores [9], differ for near-identical inputs [10], and rank features whose removal does not hurt the model [11]. Rudin [12] argues for models that are interpretable by construction. The world model and the lift model are of that kind for the plan, which lets this audit compare a built-in explanation with post-hoc ones on the same forecasts.

Explanations of time series and forecasts. Rojat et al. [13] survey explanation methods for time series. Ismail et al. [14] and Tonekaboni et al. [15] show on generated data with known important time steps that saliency methods designed for images often fail on temporal data, the approach our known-answer study takes for retail plans. Among forecasting models, the Temporal Fusion Transformer [26] reports attention weights as explanations. We know of no study that checks the explanation of a demand forecast against a measured promotional or price effect.

Promotion and price effects. Marketing science has measured promotional effects for decades: the split of a promotion's sales into brand switching, timing and quantity [32, 33], the dip after a promotion [30, 29], and the size of price elasticities and what drives it [40]. Promotion forecasting systems built on these effects include PromoCast [28]; Ma, Fildes and Huang [31] and Fildes et al. [27] review the forecasting side. These elasticities describe temporary cuts. Regular price changes move far fewer units: in Dominick's randomised store-level pricing experiments [3], regular prices were raised or lowered 10% across many categories, and volume moved about 3% over the following 16 weeks, an elasticity near -0.3. Hoch et al. [21] and Montgomery [16] model store-level price response on the same data.

Quasi-experiments in scanner data. The public Dominick's files do not mark the experimental stores or weeks. Abrams, Gui and Hortaçsu [17] reconstructed them with the original documentation and found that some published price changes did not replicate. We therefore treat the price deviations visible in the public data as a quasi-experiment, never as the experiments, and estimate their effect by difference in differences [18] with standard errors clustered by store [19, 20].

3 The explanation and the checks

3.1 Notation

A data set has series i (an item in a store) observed in periods t. The plan for series i in period t is $x_{it} = (p_{it}, \ell^1_{it}, \dots, \ell^L_{it})$: the shelf price and L lever indicators (a deal flag on Dominick's; promotion, display, feature and a tag-only price reduction flag on Frat; event and SNAP days on M5). A model fitted on the periods before an origin forecasts the next H periods from the history and the plan for those periods, $\hat{y}_{it} = f_{it}(x)$.

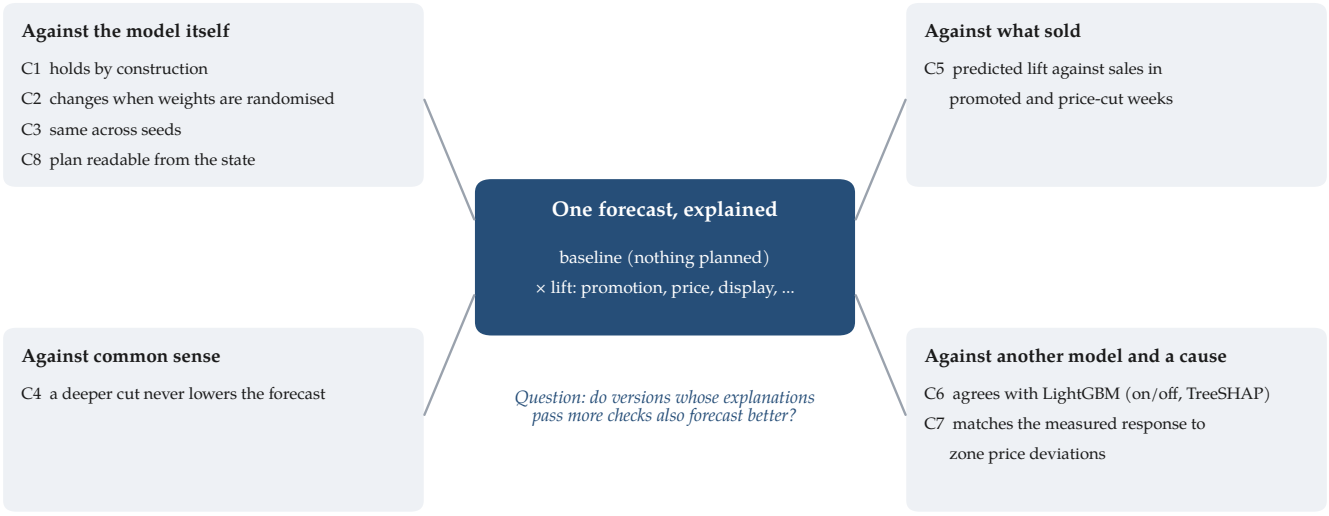


Figure 1: Each forecast's explanation is checked against the model itself, common sense, what sold, another model and a measured cause. The eight checks are fixed in `docs/PAPER2_PLAN.md` before any result; every model version and every check is reported.

The *regular price* r_i is the series' highest price over the last 12 training periods, the rule the models themselves use. The plan with *nothing planned*, x^0, sets every price to r_i and every lever to zero for all H periods. The explanation of a forecast has four parts:

$$b_{it} = f_{it}(x^0) \quad \text{baseline} \tag{1}$$

$$\Lambda_{it} = f_{it}(x) - b_{it} \quad \text{total lift} \tag{2}$$

$$c^k_{it} = f_{it}(x) - f_{it}(x \mid \ell^k = 0) \quad \text{contribution of lever } k \tag{3}$$

$$c^p_{it} = f_{it}(x) - f_{it}(x \mid p = r) \quad \text{contribution of the price} \tag{4}$$

and the interaction $\iota_{it} = \Lambda_{it} - c^p_{it} - \sum_k c^k_{it}$, which is zero only when the levers act separately. Switching a lever off switches it off for all H periods, so the dip after a promotion is part of that promotion's contribution. For the world model and the lift model the forecast is built as $\hat{y} = B + \lambda$, a baseline readout and a lift readout [34]; for LightGBM [25] the same on/off quantities are computed from its forecasts, together with TreeSHAP values ϕ^k_{it} [6] on its lever and price inputs. Unlike Shapley values, on/off contributions answer the merchant's question (what would sell without this lever) directly, and are defined for any model.

Rank agreement $\rho(a, b)$ below is the Spearman correlation over item-weeks where either a or b is non-zero, so that the many weeks with nothing planned do not inflate it.

3.2 The eight checks

C1 By construction.. With $\hat{y} = B + \lambda$, three violations, each relative to the largest forecast: the lift with nothing planned, $\max|\lambda(x^0)|$; the movement of the baseline with the plan, $\max|B(x) - B(x^0)|$; and $\max|B + \lambda - \hat{y}|$. These hold by design and are checked because a design can be broken by an implementation.

C2 Randomisation.. Following Adebayo et al. [8], the network's parts are re-initialised in turn from the output back to the input: lift readout, baseline readout, transition, encoder. After each step, the rank agreement of the total lift with the original, $\rho(\Lambda, \Lambda^{(k)})$, should fall. An explanation that survives randomisation does not depend on what the model learned.

C3 Seed stability.. Fits of the same configuration that differ only in their seed should explain a forecast alike. The score is the mean over seed pairs of $\rho(\Lambda^s, \Lambda^{s'})$, shown against the same agreement for the forecasts themselves.

C4 Price direction.. For depths $0 < 5\% < 10\% < 20\% < 30\%$ applied to the planned price, the share of item-weeks and steps in which the deeper cut gives a lower forecast than the shallower one by more than 0.1% of the forecast.

C5 Against what sold.. On item-weeks with a lever on or a price at least 5% below regular, the bias $\sum \hat{y} / \sum y$ and the rank agreement of predicted lift $\hat{y}/b$ with observed lift y/b. Merchants plan promotions for weeks they expect to sell well, so y/b overstates what the plan

caused; this check measures agreement with the data, not with a cause.

C6 Against LightGBM.. For each lever, $\rho(c^k, c^k_{\text{LGB}})$ and $\rho(c^k, \phi^k)$, on the same window. Two models fitted to the same data can agree and both be wrong; agreement is evidence of a shared signal, not of correctness.

C7 Against a measured cause.. For a spell e in which store s's regular prices sat at a log ratio δ_e from its zone peers, the model's stated response is

$$\hat{R}_e = \log \frac{\sum_{t\in E}\sum_{i\in s} f_{it}(x)}{\sum_{t\in E}\sum_{i\in s} f_{it}(x \mid p_s = p_s e^{-\delta_e})}, \qquad (5)$$

the change in the store's forecast category units if its prices had stayed with its zone, over the spell's first eight weeks E. It is compared with the measured response τ_e from difference in differences (§6): whether $\hat{R}_e$ falls inside τ_e's 95% interval, whether its sign agrees, and the pooled slope of each on δ_e.

C8 Plan read-back.. For the world model only (paper 1): the R^2 of a ridge probe that reads the day's discount back from the frozen model's state after it has rolled through the plan, against the same probe on the state before the window, which has not seen the plan.

3.3 Pass thresholds and the main test

Thresholds were fixed in the plan before any audit result was read (Table 1). For the main question, each model version (a configuration, seed and window) gets the number of checks it passes among C1 to C6, counting only those that apply to it. Within each data set the Spearman correlation across versions between that count and item-by-store WRMSSE (lower is better) is computed, and the same for each check's continuous score. A negative correlation means versions that pass more checks forecast better. The data sets are pooled by a permutation test that shuffles the scores within each data set, 10,000 times. C7 and C8 apply to a few models each and are reported per model.

4 A known-answer study

Before the checks are trusted on real data, they are run where the answer is known. Following Ismail et al. [14] and Tonekaboni et al. [15], who test time-series explanations on generated data with known important inputs, we generate weekly store data in which the effect of every part of the plan is set by us, fit correct models and models with failures planted on purpose, and ask two questions: which checks catch which failure, and do the explanations of models that pass the checks match the truth?

Table 1: Every check has a pass rule fixed before any result. Scores as defined in §3; a check that does not apply to a model (C1 and C2 for LightGBM) is not counted for it.

Check	Passes when	Applies to
C1	every violation < 0.001	lift readout models
C2	ρ after the lift readout < 0.5	neural models
C3	mean seed $\rho \geq 0.7$	configurations with ≥ 2 seeds
C4	share $< 1\%$	all
C5	bias in $[0.8, 1.25]$ and rank ≥ 0.3	all
C6	$\rho(\Lambda, \Lambda_{\text{LGB}}) \geq 0.5$	non-LightGBM

4.1 Design

The data. Three price zones of six stores sell 20 items for 260 weeks (Figure 2). Expected sales are the item's size and season times an unobserved demand shock (a first-order autoregression on the log scale, standard deviation 0.25, persistence 0.7), times the effects in Table 2. A temporary cut and a change in the regular price have different elasticities, -2.5 and -0.5, as in the literature on promotions and the Dominick's experiments. Each item gets one chain-wide regular price rise of 3% to 8%. In each zone one store's regular prices rise 10% and one store's fall 10% for 16 weeks at random times during training, and one rises and one falls starting at the forecast origin, week 253, so that every model can be scored on them exactly as on Dominick's. Sales are Poisson around their expectation.

Two worlds. In the first, cuts, promotions and displays are planned at random. In the second, merchants plan them more often in weeks when the unobserved demand shock is high (the log odds of each rise by the shock in standard deviations), as merchants do when they promote into strong weeks. Everything else is the same.

The models. In each world: the world model with three seeds; the world model fitted with each series' prices shuffled across its training weeks, so that it cannot learn a price effect (*blind to price*); the world model fitted with promotion and display shuffled the same way (*blind to levers*); LightGBM; and LightGBM blind to price. The forecast for weeks 253 to 260 is explained and every check run as on real data; the world model has the settings of the companion paper's Dominick's world model except that it is a single fit rather than an average of three, without store switching, with width 32 and 2,000 steps, to fit on a processor; LightGBM has the compan-

Table 2: The generated data have known effects for every part of the plan.

Effect on expected sales	Size
Temporary cut, elasticity on price over regular	−2.5
Regular price, elasticity on regular over list	−0.5
Promotion	+40%
Display	+25%
Promotion and display together, extra	+15%
Week after a planned week with nothing planned	−15%
Chance of a cut; depth	12%; 10% to 40%
Chance of a promotion, with a cut; without	60%; 3%
Chance of a display	8%

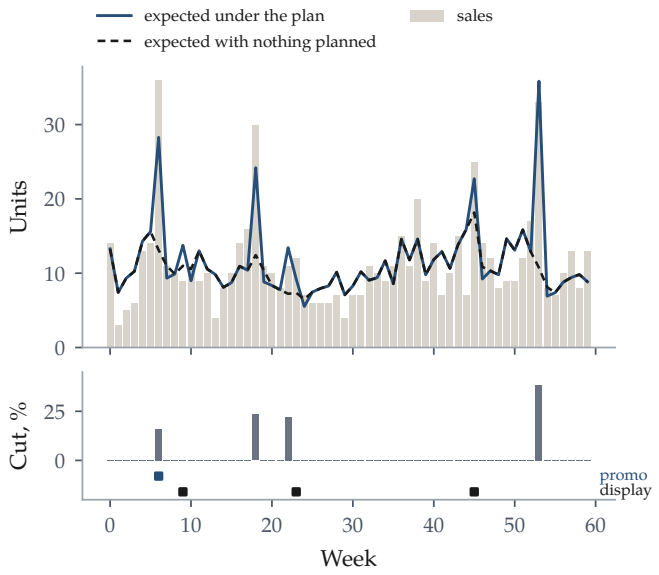


Figure 2: In the generated data the effect of every part of the plan is known. One item in one store over 60 weeks, in the world where merchants plan into high-demand weeks: sales, expected sales under the plan, and expected sales with nothing planned; below, the cut and the weeks with a promotion or a display.

ion paper's Dominick's settings. Each explanation is also scored against the truth, computed from the expected sales under the same counterfactual plans: rank agreement and the ratio of summed lifts on the 1,743 of 2,880 item-weeks with something planned, overall and by lever.

4.2 What the checks catch

Figure 3 shows every check on every model, and Table 5 in the appendix every score. Six results stand out.

Correct models pass, and their explanations are right. With plans at random, the three world-model seeds pass every check that applies except C7 and, for one seed, C2, and their lifts match the truth: rank agreement 0.99 and summed lift 1.14 times the true lift for seed 1, with the price, promotion and display parts each within 13% of the truth. Seeds agree on their lifts as closely as on their forecasts (C3, 0.99). LightGBM's lifts rank 0.85 with the truth at 1.09 times.

The price check catches a model blind to price, and flags LightGBM too; the check against sales does not. All four models fitted without a usable price fail C4: 31.5% and 27.2% of item-weeks for the world model, 49.5% and 43.4% for LightGBM, against 0% for the world models fitted on real prices. It also fails both LightGBM models fitted on real prices (7.6% and 9.9%), whose lifts are close to the truth and whose regular-price response passes C7: trees respond to price in steps, so a deeper cut sometimes lowers the forecast. C4 is sensitive to a model ignoring price but not specific to it. All four pass C5: their forecasts on planned weeks sum to 0.88 to 0.93 of sales, with rank agreement 0.40 to 0.59. Their explanations are wrong in a way a merchant would act on: the price part is zero, and the promotion, which usually comes with a cut, takes its place, at 1.95 to 2.37 times its true lift across the four (Figure 4). Both price-blind world models also fail C1 narrowly, because the audit takes the regular price from the true history while these models learned it from a shuffled one; that is a property of how the failure was planted, not a sign that C1 detects it.

Only a per-lever comparison catches lifts given to the wrong lever. The per-lever comparison was added after the first results, without a pass rule fixed in advance, and is exploratory. The world model fitted without usable promotion and display flags passes C1, C4 and C5, and its total lift agrees with LightGBM's (C6, 0.67 and 0.54) and ranks 0.89 with the truth. Its explanation gives the promotion 0.14 and the display 0.05 times their true lift, and the price 1.35 times. Every check on the total lift except C2 passes it. Its per-lever agreement with LightGBM does not: 0.29 for the promotion and −0.17 for the display, against 0.80 for the correct world model.

When plans follow demand, every check passes and the lifts are a third too large. In the second world the three world-model seeds pass C1, C3, C4, C5 and C6, and their forecasts are more accurate than in the first (item-by-store RMSSE 0.345 against 0.401): a plan made in high-demand weeks tells the model about the demand. Their lifts are 1.33 to 1.40 times the true lift, and the display's 1.81 to 1.93 times. C5 cannot see this: sales in planned weeks are high for the same reason, and the forecasts sum to 1.06 times them. C6 cannot see it either: LightGBM, fitted to the same data, overstates by 1.26 times, and the two agree at 0.75. No check that com-

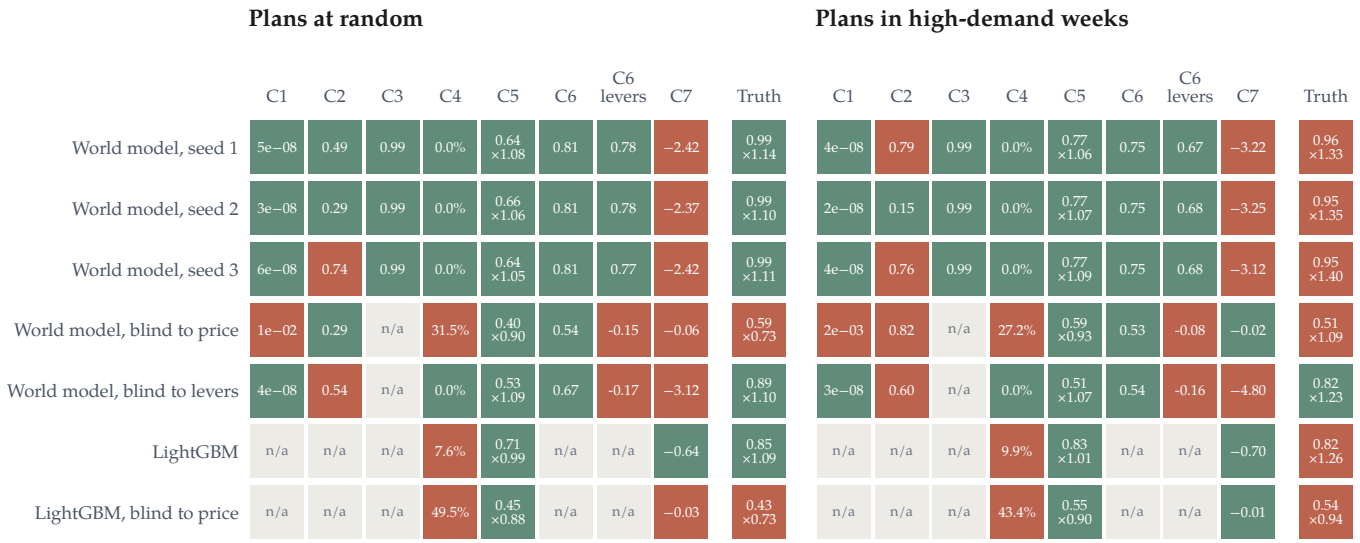


Figure 3: Correct models pass the checks; so do models whose explanations are wrong in ways only some checks can see. Each check's score and its pass (green) or fail (red) under the rules fixed in advance, for every model in both worlds; grey, does not apply. C5: rank of predicted with observed lift, and summed forecast over summed sales. C6 levers: the lowest per-lever rank agreement with LightGBM, reported but without a rule fixed in advance. C7: pooled slope of the stated response to the price spells at the origin, passed when inside the 95% interval of the measured slope. Truth: rank agreement of the model's lifts with the true lifts and the ratio of their sums, green when the rank is at least 0.7 and the ratio within 0.8 to 1.25.

pares the explanation with the model, with the data or with another model fitted to the same data can find a bias that the data themselves carry.

The world model treats a regular price change like a temporary cut. For the price spells that start at the origin, the world model's stated response is -2.42 to -2.37 per unit of log price across seeds with plans at random and -3.25 to -3.12 when plans follow demand, 4.7 to 6.5 times the true -0.5; and -2.5 is the temporary-cut elasticity (Figure 5). It has learned how sales answer a cut and applies it to a change in the regular price. LightGBM, whose inputs include the item's recent prices, states -0.64 and -0.70. The measured response from difference in differences on the generated sales is -0.73 (-1.41 to -0.06) and -0.74 (-1.90 to 0.41): imprecise with six spells and three or four controls each, as on Dominick's, but enough to reject the world model's answer in both worlds and not LightGBM's. In the second world the measured interval also contains zero, so there C7 passes the price-blind models too: with six spells it can reject a response five times too large, not one that is absent. Apart from C2 for one seed, C7 is the only check the world model fails in the first world, and the only one that finds this error.

Randomisation does not separate working models from broken ones. After the lift readout is re-initialised, the rank agreement of the lifts with the original is 0.15 to 0.82 across world models, and with every part re-initialised it is 0.44 to 0.70 (Figure 6). The construction itself ties the lifts to the weeks with something planned and to the size of the item, so a network with random weights still ranks them much as the trained one does. C2 fails correct seeds and passes broken models; as specified, it does not test a lift built into the model, and we do not use it as evidence below.

What the study says about the checks. Across the 14 models, the share of checks passed tracks how well a model ranks the true lifts (Spearman 0.87) but not how close its lifts are in size (-0.27 with the negative absolute log ratio): the models that pass the most checks include the confounded ones whose lifts are a third too large. Accuracy tracks the gross failures, the price-blind models being the least accurate, and misses the subtle ones. The checks earn their place as follows. C4 finds a model that ignores price, and also flags one whose price response is merely uneven. Per-lever agreement with another model finds lifts given to the wrong lever. C1 and C3 confirm construction and stability. C5 and C6 confirm agreement but cannot find a bias the data carry, and C2 as specified finds nothing. Only a measured cause, C7, finds that a model's stated response to a price change is wrong, and only one kind of response, a sustained regular-price change, has a measured cause on public retail data.

5 Data, model versions and protocol

Data. The three public data sets of the companion paper [34]: M5, daily Walmart sales of 3,049 items in 10

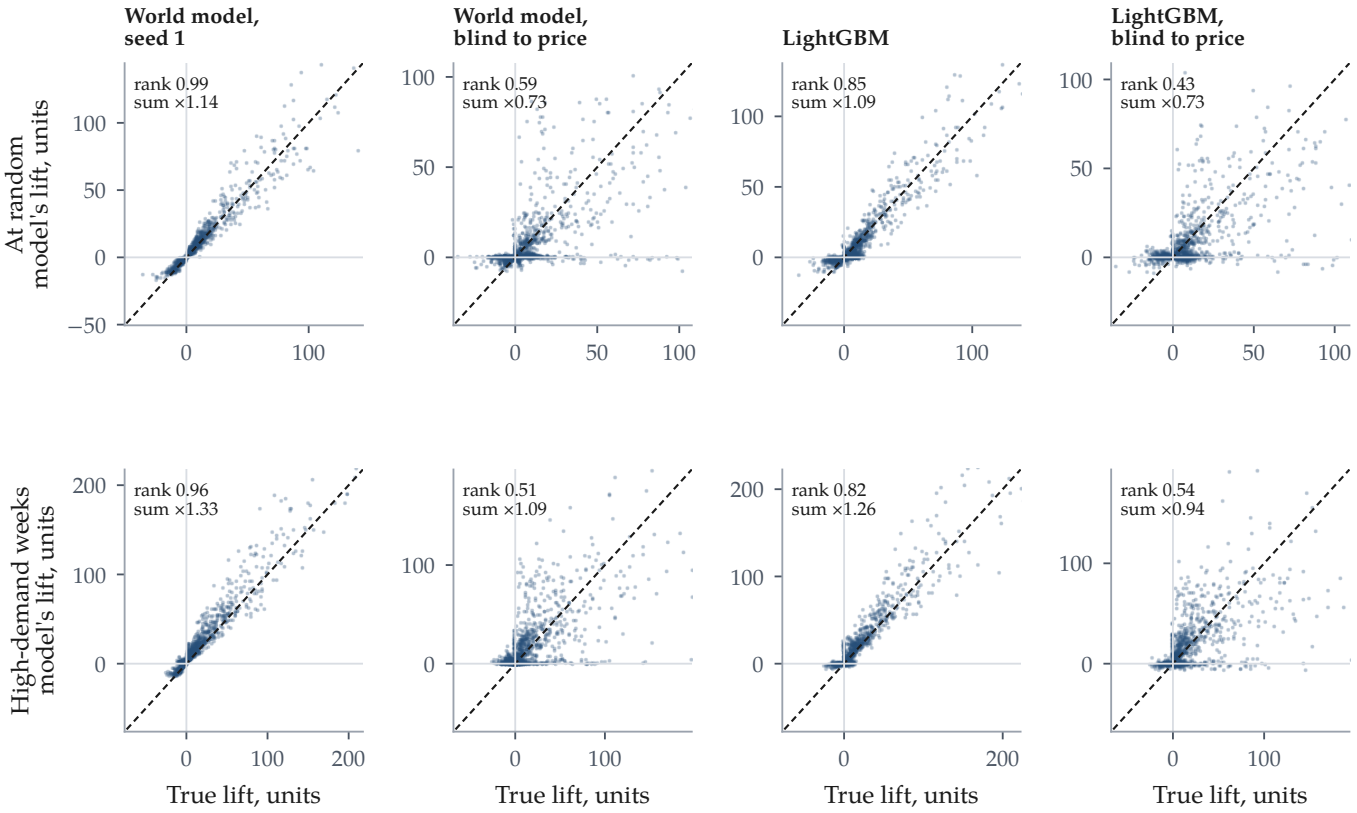


Figure 4: When plans follow demand, both the world model and LightGBM overstate what the plan adds, and agree with each other. Each point is an item-week with something planned: the model's total lift against the true lift, units. Dashed: equal. Top, plans at random; bottom, plans made more often in high-demand weeks.

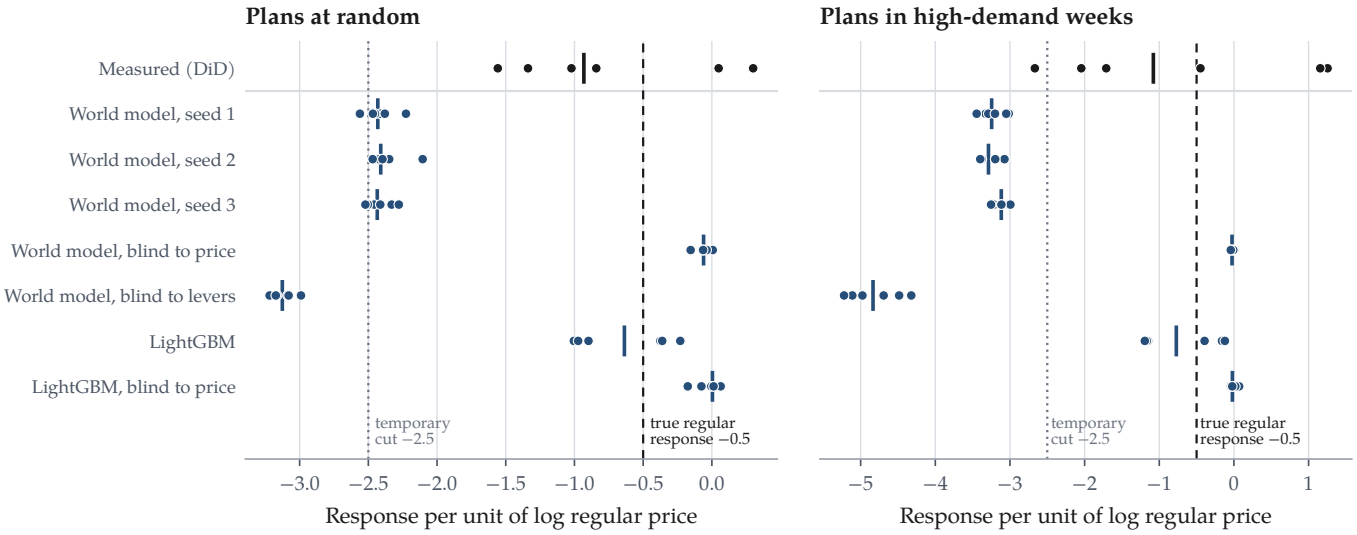


Figure 5: Every world model states the response to a regular price change as if it were a temporary cut; LightGBM does not. The stated change in store category units per unit of log price for each of the six price spells that start at the forecast origin (dots; bar, median), against the measured response from difference in differences on the generated sales (top row), the true response (-0.5, dashed) and the temporary-cut elasticity (-2.5, dotted).

stores from 2011 to 2016, with prices, event days and SNAP days [24]; Dominick's Finer Foods, weekly scanner data by store for a Chicago chain from 1989 to 1997, with price, a promotion code and cost [22], of which the companion paper's six final categories are audited (cheese, snack crackers, soft drinks, bottled juice, analgesics, frozen entrees); and Breakfast at the Frat, weekly sales of 55 breakfast items in 77 stores from 2009 to 2012, with display, feature and tag-only price reduction flags [23]. No raw data is redistributed.

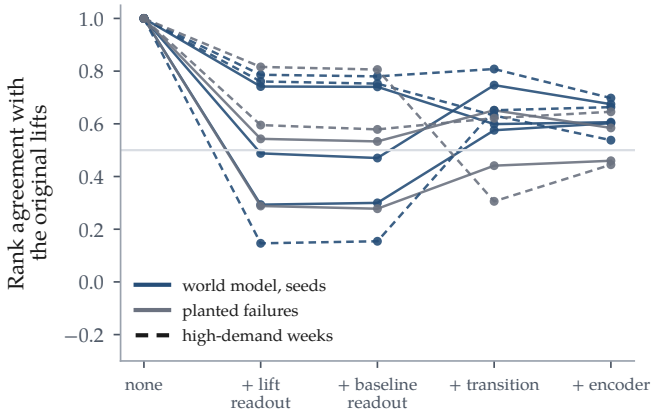


Figure 6: Re-initialising the network leaves the lifts' ranking largely in place. C2 for every world model, both worlds: rank agreement of the total lift with the original after the lift readout, then the baseline readout, the transition and the encoder are re-initialised in turn.

Model versions. Every saved world-model and lift-model checkpoint from the companion paper is audited: the development and final windows of Frat, the development versions on Dominick's, and the M5 practice, confirmation and test windows. The Dominick's final world models, each an average of three fits, were not saved in the companion paper; they are refitted with settings unchanged for the audit, three seeds each, and the companion paper keeps the scores of its original runs. Each version's accuracy is read from its own run record. LightGBM is refitted on the Dominick's and Frat final windows and the M5 test month for C6, with the settings of the companion paper.

Protocol. The audit reads models and never tunes them. Every check runs on every version it applies to and every result is reported, including failures. The plan, its thresholds and every amendment are dated in `docs/PAPER2_PLAN.md`; each amendment was made before the results it affects were read.

6 A ground truth from zone price deviations

6.1 Finding the deviations

Dominick's set regular prices by zone: within a zone every store was meant to carry the same regular price [22]. For each store and item we take the log of the week's regular price (no deal code, not more than 5% below the item's highest price in that store over the last 12 weeks) over that item's median regular price in that store over all weeks, and subtract the median of the same across the zone's stores that week. Moves made zone-wide cancel, and so do an item's usual price differences between stores; a store moved on its own stands out. An episode is at least 12 consecutive weeks in which this quantity is at least 3% in the same direction for at least 60% of the store's items, with a median of at least 5% over the episode. Controls are stores of the same zone within 1% of their zone over those weeks. An episode with fewer than three controls, or without the 26 comparison weeks before its start, is listed but not scored.

The detector was written, and its episodes listed, before any model was scored against them. A first version measured stores against the zone's most common price and found one episode, in analgesics: in the experiments a third of a zone's stores moved up and a third down, so the most common price was often a moved one. The change is recorded, with its date, in the plan.

6.2 Measuring their effect

For episode e of store s with controls $\mathcal{C}_e$, the response is the difference in differences on log category units,

$$\log Y_{jw} = \alpha_j + \gamma_w + \tau_e D_{jw} + \varepsilon_{jw}, \qquad j \in \{s\} \cup \mathcal{C}_e, \quad (6)$$

over the 26 weeks before the episode and its first eight weeks, where Y_{jw} is store j's category units in week w and D_{jw} marks the treated store in the episode weeks. Standard errors are clustered by store [19, 20] with the CR1 correction and t critical values on stores minus one degrees of freedom; with 4 to 25 stores per episode, single intervals are wide by construction. The plan scores each episode by its own interval; after the placebo showed those intervals to be too narrow, the pooled slope of τ_e on δ_e across episodes, with an interval robust to unequal variances, was added as the summary used for C7. The estimate over the full episode is kept in the list as well. The models forecast at most eight weeks ahead, so each model is fitted on the data up to the week before an episode and scored on its first eight weeks, the same weeks as τ_e.

6.3 What the deviations show

The detector finds 54 episodes in the six categories, 38 with at least three control stores (Figure 7). Most lie between Dominick's weeks 190 and 240, April 1993 to April 1994. Hoch, Drèze and Purk published their experiments in 1994, and the public files do not date them, so whether these spells are those experiments cannot be checked. The scored stores' prices sit 6% to 10% off their zone peers. Over the first eight weeks of each scored episode, category units move against price in 63% of episodes. The pooled slope of the measured response on the price change is -0.33, with a 95% interval of -0.68 to 0.01 (heteroskedasticity-robust across episodes), and over the full episode -0.52 (-0.81 to -0.23). Both are

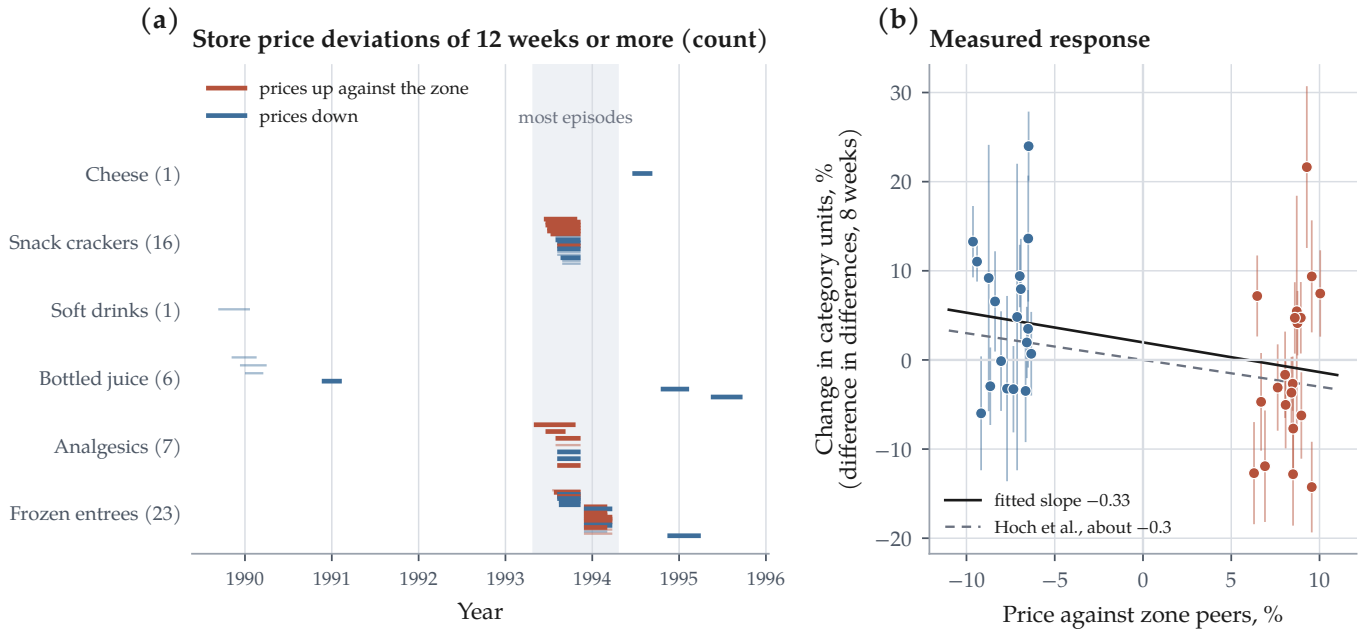


Figure 7: The price deviations cluster between spring 1993 and spring 1994, and on average sales move against price. (**a**) Every episode found in the six categories (count in brackets); faint: fewer than three control stores, listed but not scored. (**b**) For the 38 scored episodes, the store's price against its zone peers and the measured change in its category units over the episode's first 8 weeks (difference in differences against same-zone stores, store and week fixed effects, 95% interval clustered by store; these intervals are too narrow, see Figure 8). Solid line: fitted slope; dashed: the elasticity near −0.3 implied by the experiments [3].

in line with the elasticity near −0.3 implied by Hoch, Drèze and Purk's report of a 3% volume change for a 10% price change, and neither is precise. These intervals across episodes were added after the placebo was seen (below).

Three robustness analyses, fixed in advance, and two added after seeing them qualify this ground truth (Figure 8).

Prices move before the detected start. The detector dates an episode from the first week in which most of the store's items are at least 3% off. Prices ease in before that: in the eight weeks before the detected start, 47% of store-weeks are already 3% or more off in the episode's direction (panel a). Part of the 26 comparison weeks is therefore treated, which biases the 8-week estimate towards zero and shows in the event study as a gradual decline in signed sales over the weeks before the start (panel b). The full-episode estimate, which compares the whole spell, is larger, −0.52 against −0.33.

Single episodes are noise. Moved 26 weeks earlier, to weeks where prices had not moved, the same estimator gives a pooled slope of −0.26 (−0.72 to 0.19), and 42% of the 38 placebo intervals exclude zero, where 5% should. The intervals clustered by store are far too narrow: their median standard error is 0.02 while placebo estimates spread with a standard deviation of 0.12. With few stores per episode, weekly category sales that move together over many weeks, and errors clustered by so few stores, this is the failure Bertrand, Duflo and Mullainathan [19] and Cameron and Miller [20] describe. A single episode's interval, and with it the plan's per-episode score for C7, cannot be trusted; the comparison of models with the measured response is made on the pooled slope and its interval.

The pooled response is stable but imprecise. Over the nine detector settings (Table 3) the number of scored episodes ranges from 26 to 45 and the pooled slope from −0.38 to −0.16. By category, the slope is negative in 3 of the 3 categories with at least five episodes, and every category's interval is wide (panel d).

What survives is a pooled response of store category units to a sustained change in regular prices of −0.33 over the first eight weeks, whose interval just includes zero and overlaps the placebo's, and −0.52 over the full spell, whose interval excludes zero. The 8-week figure is pulled towards zero by prices that moved before the detected start. That is enough to tell a model that treats a regular price change like a temporary cut, an elasticity several times larger, from one that does not; it is not enough to tell a model with no price response from a correct one. That is the comparison C7 can make on public data.

What this ground truth is, and is not. These deviations may include zone moves that were not experiments, and

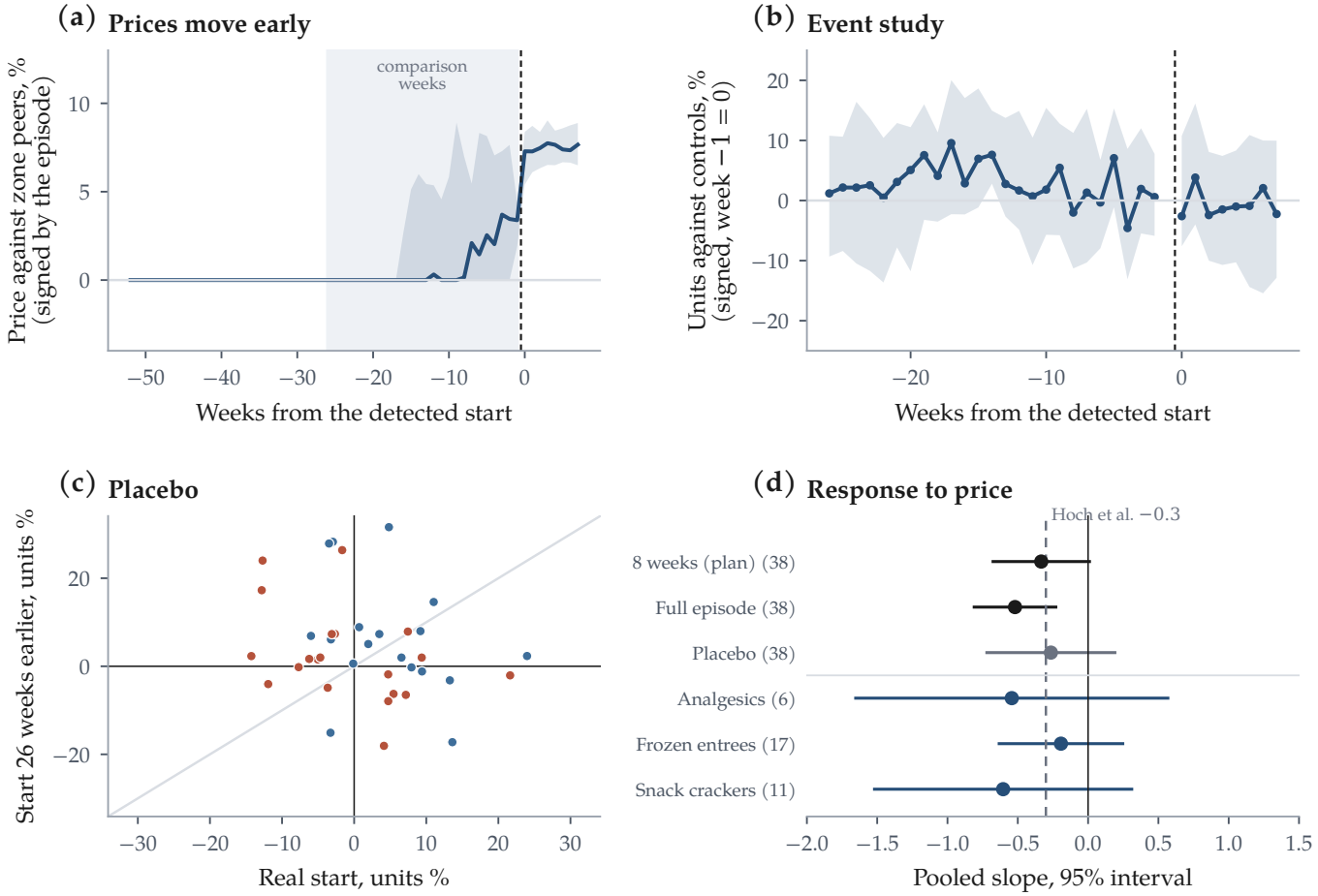


Figure 8: Prices start moving before the detected start, single episodes are mostly noise, and only the pooled response carries information. (**a**) The store's regular prices against its zone peers, signed by the episode's direction: median and interquartile band across the 38 scored episodes; shaded, the 26 comparison weeks of the difference in differences. (**b**) Event study: weekly coefficients signed the same way, week −1 as reference; median and interquartile band. (**c**) The 8-week estimate with the start moved 26 weeks earlier, where prices had not moved, against the real estimate; red, prices up; blue, down. (**d**) Pooled slopes of the response on the price change with 95% intervals, for the plan's 8-week estimate, the full episode, the placebo, and each category with at least five episodes (count in brackets). The detection rule, controls and estimator were fixed in advance; panels (a) and (d) were added after the placebo was seen.

Table 3: The pooled response keeps its sign under every detector setting. Episodes found and scored, and the pooled 8-week slope, for each weekly size and episode median; the plan's setting in bold.

Each week	Episode median	Episodes	Scored	Slope
2%	4%	64	45	−0.38
2%	5%	55	38	−0.37
2%	7%	38	26	−0.20
3%	4%	63	45	−0.32
3%	**5%**	**54**	**38**	−0.33
3%	7%	37	26	−0.16
4%	4%	59	43	−0.32
4%	5%	54	38	−0.34
4%	7%	37	26	−0.17

the models were trained on data that include earlier deviations. The ground truth is the response of a store's category units to a sustained change in its regular prices relative to its zone, over eight weeks. It is not the response to a temporary cut, which the companion paper's models mostly learn from, and a model can be right about one and wrong about the other. The known-answer study (§4) has both, with different true elasticities, for that reason.

7 The audit of the saved model versions

This section is filled from `results/paper2/summary/` *when the audit of the saved versions finishes on Colab: the pass counts by check and data set, every check across versions, the models' stated response to the zone price deviations against the measured one, and the main test. The figure and table code is in* `paper2/figs/real.py` *and has been run end to end.*

8 Discussion

Faithful is not correct, in retail too. The world model's explanation is faithful by construction: the baseline does not move with the plan, the lift is zero when nothing is planned, and the two add up to the forecast. In the known-answer study that explanation was right about what each lever adds when plans were made at random, and wrong in two ways that no check against the model could see. When merchants plan into strong weeks, the lifts are a third too large, because the data tie the plan to demand the model cannot see. And asked about a change in the regular price, the model answers with the response to a temporary cut, about five to six and a half times the truth. Molecular attributions in MolSanity [2] came apart from the truth for a similar reason: a model can be faithful to what it learned and what it learned can be wrong.

What this means for a merchant reading a lift. A lift read from a model fitted to a retailer's own history inherits that retailer's planning habits. If promotions go into weeks that would sell well anyway, the model's promotion lift overstates what the promotion adds, and an accurate forecast is no protection: in the known-answer study the correct world-model seeds and LightGBM forecast better when plans followed demand than when they were made at random. The check against what sold, the one most often run in practice, agrees with the overstated lift for the same reason. Two things help. Agreement between models fitted to the same data should not be read as confirmation. And a model's answer to a question it was not trained on, here a lasting change in the regular price from a model that mostly saw temporary cuts, should be checked against a measured response before it is used to set prices.

Which checks to run. The known-answer study ranks the checks. The price-direction check (C4) is cheap and caught every model blind to price, though it also flags LightGBM, whose response to price comes in steps. Per-lever agreement with a second model, an exploratory comparison, caught lifts given to the wrong lever, which every check on the total lift missed. Construction (C1) and seed stability (C3) are worth confirming once. The check against what sold (C5) and agreement with LightGBM on the total lift (C6) confirm agreement, not correctness. Randomisation (C2), the standard sanity check for attributions [8], does not test an explanation built into the model: re-initialising every part left the lifts' ranking largely in place. A version on the lift as a share of the baseline, or within planned weeks only, may work; we did not test it. A measured cause (C7) is the only check that found the regular-price error, though with six spells it could not tell a missing price response from a correct one. We expect that a measured promotion effect would be needed to find the bias from planning into strong weeks; we did not test this, and public retail data offer none.

Ground truth on public data is scarce and noisy. Dominick's zone price deviations give a pooled response in line with the one the chain's own experiments reported, but single episodes are noise, intervals clustered by a handful of stores are far too narrow, and prices start moving before the detector sees them. A retailer running its own tests would do better with the design under its control: stores assigned at random, a known start date, and enough stores per arm for honest intervals.

The audit of the saved versions. *To be written with the results of §7: whether the real versions behave like the known-answer models, and whether versions that pass more checks forecast better.*

9 Conclusion

Plan-lift explanations built into a forecasting model can be faithful, stable and in agreement with sales and with another model, and still be wrong about what a plan causes. In generated store data with known effects, the checks that compare an explanation with the model, the data or another model caught a model blind to price and, compared lever by lever, lifts given to the wrong lever. They did not catch lifts inflated by planning into strong weeks, or a model that answers a question about the regular price with the response to a temporary cut. Only a measured cause found the second. On public retail data the one such cause we could measure, Dominick's zone price deviations, is real but imprecise. *The audit of the companion paper's saved versions, and the test of whether faithful explanations go with accurate forecasts on real data, complete this section when the Colab run finishes.*

10 Limitations and threats to validity

The ground truth covers one effect on one data set. C7 measures the response of a store's category units to a sustained change in its regular prices, on Dominick's. It says nothing directly about promotions, displays or temporary cuts, or about M5 and Frat, where no such variation is used. A model can be right about regular prices and wrong about promotions; the known-answer study shows how far the other checks go in covering the rest.

The deviations are inferred. The public files do not mark the experimental stores or weeks. Some deviations

may be zone moves made for other reasons, and a store moved for a reason that also moves its sales would bias τ_e. The event study and the placebo address this for trends that start before the price moves; they cannot rule out a shock that starts in the same week.

Few control stores. Zones have few stores, so each scored episode has 3 to 24 controls. Intervals clustered by store with so few clusters are wide and only approximately calibrated; the pooled slope across episodes carries the comparison.

The baseline is a choice. Every lift is measured against nothing planned, with the regular price taken as the highest of the last 12 periods. A different rule gives different baselines and lifts. The rule is the models' own and is applied the same way to every model and to the truth in the known-answer study.

The versions are not a designed sample. The audited versions are what the companion paper saved. They share data and differ in many settings at once, so the correlation across versions is descriptive; it is computed within data set and tested by permutation within data set, but versions are not independent draws.

Refitted versions. The Dominick's final world models are refits with unchanged settings, not the runs whose scores the companion paper reports; seed-to-seed differences between the two are expected and are measured by C3.

The generated world is simpler than retail. The known-answer data are multiplicative in the plan, with one shock process and no cannibalisation between items. They test whether a check can catch a failure, not how often the failure occurs in practice.

11 Reproducibility, compute and licence

Every number, table and figure in this paper is generated from committed records by `make -C paper2`, and a continuous integration job rebuilds the paper from a fresh checkout on every change and fails on any LaTeX error, undefined reference or missing font. The checks are in `src/rwm/audit/`; the zone detector and estimators in `src/rwm/audit/zones.py`; the generated data in `src/rwm/data/synthetic_levers.py`. The runs:

`scripts/zone_episodes.py` lists the episodes and their responses (`results/paper2/zone_episodes.csv`);

`scripts/zone_robustness.py` the event study, placebo and sensitivity;

`scripts/known_answer.py` the known-answer study;

`scripts/audit_runs.py` and `scripts/paper2_colab.sh` the audit of the saved versions, C6 and C7;

`scripts/audit_summary.py` the tables committed under `results/paper2/`.

Compute. The known-answer study fits 14 models on a two-core processor: each world model in 5 to 23 minutes (69 minutes in all, the longest while another job shared the processor), each LightGBM in seconds. The zone episode list, event study, placebo, sensitivity and price path read only the data and take minutes on the same processor. The audit of the saved versions runs on a Colab GPU; its run times are recorded with its results.

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Data availability. M5 is available from Kaggle [24], Dominick's from the Kilts Center for Marketing at the University of Chicago Booth School of Business [22], and Breakfast at the Frat from dunnhumby [23], each under its own terms; the repository fetches them and redistributes none.

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A Every zone price episode

Table 4: All 54 episodes the detector finds, with their measured response. Weeks: Dominick's week numbers of the episode. Price: the store's median log price against its zone peers over the episode, %. Controls: same-zone stores within 1% of their zone. 8 weeks: difference-in-differences estimate on log category units over the first eight weeks, %, with its interval clustered by store (too narrow, §6); Full: over the whole episode. * Fewer than three controls: listed, not scored.

Category	Zone	Store	Weeks	Price	Controls	8 weeks	95% interval	Full
Analgesics	2	50	190–215	8.8	16	4.1	0.6 to 7.6	−5.7
Analgesics	2	49	197–209	8.5	16	−2.7	−5.6 to 0.3	−3.2
Analgesics	2	48	203–218	8.1	11	−5.0	−9.8 to −0.2	−7.1
Analgesics	7	109	203–218	6.0	2	8.2	−4.1 to 20.4	9.1 *
Analgesics	1	32	204–218	−6.5	7	13.6	6.7 to 20.6	14.9
Analgesics	2	88	204–218	−6.7	10	−3.5	−9.1 to 2.2	−2.0
Analgesics	2	89	204–218	6.3	10	−12.7	−18.3 to −7.1	−9.4
Bottled juice	12	105	9–24	−7.9	7	−7.2	−18.9 to 4.4	−3.2 *
Bottled juice	1	123	14–30	−5.5	11	7.8	3.6 to 12.1	11.5 *
Bottled juice	6	40	17–28	−6.6	6	8.2	−9.0 to 25.3	5.8 *
Bottled juice	6	40	63–75	−7.7	6	−3.2	−13.5 to 7.1	−0.1
Bottled juice	2	84	266–283	−9.4	24	11.0	8.9 to 13.2	12.7
Bottled juice	2	84	296–315	−9.2	22	−6.0	−12.3 to 0.3	−0.9
Cheese	6	77	249–261	−7.1	7	4.8	−12.3 to 21.9	4.1
Frozen entrees	5	8	201–218	10.2	1	−26.8	−26.8 to −26.8	−31.0 *
Frozen entrees	2	47	202–218	6.7	9	−4.7	−10.1 to 0.7	−3.2
Frozen entrees	2	28	204–218	−6.9	9	8.0	2.4 to 13.5	7.9
Frozen entrees	1	71	204–218	9.3	4	21.6	12.7 to 30.6	17.1
Frozen entrees	2	91	204–218	−8.4	9	6.6	1.0 to 12.1	9.7
Frozen entrees	6	40	205–218	−7.6	2	−3.2	−6.8 to 0.4	−2.1 *
Frozen entrees	12	114	205–218	−8.7	3	9.2	−5.7 to 24.0	7.5
Frozen entrees	2	119	205–218	−8.0	9	−0.1	−5.6 to 5.4	−1.1
Frozen entrees	1	2	220–237	9.0	4	−6.2	−11.0 to −1.5	−12.8
Frozen entrees	5	8	220–237	9.8	2	−13.0	−95.9 to 69.9	−7.7 *
Frozen entrees	2	9	220–237	−9.6	9	13.3	9.4 to 17.2	7.8
Frozen entrees	7	33	220–237	5.9	2	2.3	1.0 to 3.6	−1.1 *
Frozen entrees	6	40	220–237	−10.9	1	21.0	21.0 to 21.0	15.6 *
Frozen entrees	2	49	220–234	8.9	9	4.7	0.8 to 8.6	5.0
Frozen entrees	1	68	220–237	7.6	4	−3.1	−7.8 to 1.7	−12.2
Frozen entrees	1	71	220–237	10.0	4	7.4	2.7 to 12.2	−5.4
Frozen entrees	7	75	220–234	−10.2	2	−0.3	−1.6 to 1.0	−4.7 *
Frozen entrees	2	76	220–234	8.6	9	4.7	0.8 to 8.6	4.8
Frozen entrees	1	95	220–234	8.1	4	−1.7	−6.4 to 3.1	−6.4
Frozen entrees	2	110	220–234	8.4	9	−3.7	−7.6 to 0.2	−1.6
Frozen entrees	2	119	220–237	−6.6	9	2.0	−1.9 to 5.9	−2.8
Frozen entrees	1	130	220–233	−7.3	4	−3.3	−8.0 to 1.5	2.1
Frozen entrees	1	71	270–290	−7.0	14	9.4	6.0 to 12.8	5.0
Soft drinks	12	105	1–20	−7.4	7	18.0	18.0 to 18.0	8.8 *
Snack crackers	2	50	196–216	9.6	10	9.4	3.2 to 15.6	−13.1
Snack crackers	2	48	197–218	8.5	9	−12.8	−18.5 to −7.1	−13.2
Snack crackers	2	76	197–218	6.9	9	−11.9	−18.1 to −5.8	−6.8
Snack crackers	2	49	198–218	9.6	9	−14.3	−19.2 to −9.3	−6.6
Snack crackers	2	89	198–218	8.5	9	−7.7	−12.6 to −2.8	−1.1
Snack crackers	2	47	200–218	6.5	9	7.2	2.7 to 11.6	9.6
Snack crackers	2	28	203–218	−6.5	9	24.0	20.2 to 27.8	22.4
Snack crackers	5	94	203–218	−7.1	2	−1.2	−8.0 to 5.7	−6.2 *
Snack crackers	1	2	204–218	8.7	4	5.5	−7.4 to 18.3	2.9
Snack crackers	7	75	204–218	−7.4	2	−9.9	−10.9 to −8.8	−12.6 *
Snack crackers	2	88	204–218	−8.6	9	−2.9	−7.2 to 1.3	−0.6
Snack crackers	12	114	204–218	−6.7	2	9.2	−2.7 to 21.0	6.0 *
Snack crackers	2	119	204–218	−6.5	9	3.5	−0.8 to 7.7	4.1
Snack crackers	2	113	206–218	−6.3	9	0.7	−3.9 to 5.3	0.1
Snack crackers	6	40	207–218	−8.7	2	9.5	−18.8 to 37.8	14.6 *
Snack crackers	12	129	207–218	−7.4	2	2.1	−4.9 to 9.0	−0.9 *

B The plan and its amendments

The plan (`docs/PAPER2_PLAN.md`) was committed before any audit result on 13 October 2026. Four dated amendments followed, each made before the results it affects were read:

1. Before any model lift: the episode detector measures each store against its zone peers rather than the zone's most common price.
2. After the episode list, before any model lift: at least three controls to score an episode; models scored on an episode's first eight weeks, against an 8-week difference in differences.
3. Before any audit result: the known-answer study, the C7 robustness analyses, the pass thresholds and the main test.

4. After the C7 robustness results, before any model lift for C7: the price path and the pooled slope intervals, labelled exploratory, and the comparison of models with the measured response made on the pooled slope.

C Every known-answer model

Table 5: Every model in the known-answer study, with its explanation scored against the truth. Lift against truth: rank agreement and ratio of summed lifts on item-weeks with something planned, overall, and the ratio for the price part. C4: item-weeks lower after a deeper cut. C5: summed forecast over summed sales on planned weeks. Passed: checks passed of those that apply (C1 to C6). RMSSE: item by store, weeks 253 to 260.

	Lift against truth						
Model	rank	sum	price	C4	C5	Passed	RMSSE
Plans at random							
World model, seed 1	0.99	1.14	1.13	0.0%	1.08	6/6	0.401
World model, seed 2	0.99	1.10	1.09	0.0%	1.06	6/6	0.401
World model, seed 3	0.99	1.11	1.11	0.0%	1.05	5/6	0.398
World model, blind to price	0.59	0.73	0.01	31.5%	0.90	3/5	0.510
World model, blind to levers	0.89	1.10	1.35	0.0%	1.09	4/5	0.450
LightGBM	0.85	1.09	1.03	7.6%	0.99	1/2	0.389
LightGBM, blind to price	0.43	0.73	−0.00	49.5%	0.88	1/2	0.515
Plans in high-demand weeks							
World model, seed 1	0.96	1.33	1.22	0.0%	1.06	5/6	0.345
World model, seed 2	0.95	1.35	1.24	0.0%	1.07	6/6	0.349
World model, seed 3	0.95	1.40	1.27	0.0%	1.09	5/6	0.344
World model, blind to price	0.51	1.09	−0.00	27.2%	0.93	2/5	0.489
World model, blind to levers	0.82	1.23	1.63	0.0%	1.07	4/5	0.480
LightGBM	0.82	1.26	1.13	9.9%	1.01	1/2	0.322
LightGBM, blind to price	0.54	0.94	−0.02	43.4%	0.90	1/2	0.497